# Online Resource 1

**Article:** Calibration-guided robust reallocation of emergency resources under evolving flood risk forecasts: a digital twin framework. **Journal:** *Natural Hazards*. **Authors:** Dannuri Mounika and Johnson Kolluri, School of Computer Science & Artificial Intelligence, SR University, Warangal 506371, Telangana, India. **Corresponding author:** Johnson Kolluri, johnson.kolluri@sru.edu.in.

Contents: S1 experiment configuration; S2 allocation results with interquartile ranges (2018, 2019); S3 sensitivity to $\alpha$ and $\eta$; S4 alternative need-share assumptions; S5 run log with input checksums. Code and notebooks: https://github.com/dmounikaresearch-cmyk/kerala-flood-cgrr.

## S1 Experiment configuration

| Parameter | Value |
|---|---|
| smoke_test | False |
| events | ['2018', '2019'] |
| model.alpha | 0.1 |
| model.eta | 0.1 |
| model.beta | 0.5 |
| model.tau_max | 12.0 |
| model.lam | 0.01 |
| model.mu | 0.05 |
| model.rho | 0.5 |
| model.fixed_width | 0.3 |
| model.b4_gamma_frac | 0.5 |
| model.need_beta | [2.0, 98.0] |
| model.resources.team.kappa | 150 |
| model.resources.team.hmax | 0.3 |
| model.resources.team.mode | road |
| model.resources.boat.kappa | 60 |
| model.resources.boat.hmax | 99.0 |
| model.resources.boat.mode | boat |
| model.resources.medical.kappa | 80 |
| model.resources.medical.hmax | 0.3 |
| model.resources.medical.mode | road |
| experiment.n_runs | 200 |
| experiment.n_scen_b3 | 30 |
| experiment.b5_search | 60 |
| experiment.seed | 42 |
| experiment.sensitivity_alpha | [0.05, 0.1, 0.2, 0.3] |
| experiment.sensitivity_eta | [0.0, 0.05, 0.1, 0.2, 0.3] |
| experiment.demand_means | [0.01, 0.02, 0.05] |

Values marked ASSUMED in the configuration file: $\beta$, $\tau^{\max}$, need share Beta(2, 98), resource capacities $\kappa_r$ and depth limits $h_r^{\max}$. Stocks: 2018, 55 teams, 110 boats, 16 medical units; 2019, 9 teams, 18 boats, 8 medical units, across the hubs Thrissur, Ernakulam, Thiruvananthapuram and Kozhikode.

## S2 Allocation results, 2018 (median [interquartile range] over 200 runs)

| Method | Worst-unit weighted unmet | Total unmet (people) | Mean response (h) | P95 response (h) | Coverage (%) |
|---|---|---|---|---|---|
| B1 | 138,870 [89,500, 187,000] | 281,293 [198,000, 387,000] | 0.78 [0.64, 0.88] | 1.48 [1.45, 1.67] | 36.3 [28.6, 42.2] |
| B2 | 82,750 [54,500, 147,000] | 159,526 [99,500, 253,000] | 1.11 [0.89, 1.23] | 2.75 [2.09, 2.81] | 63.0 [53.5, 71.4] |
| B3 | 53,543 [15,000, 117,000] | 69,147 [22,600, 150,000] | 0.96 [0.75, 1.09] | 2.08 [2.08, 2.08] | 85.2 [73.9, 94.2] |
| B4 | 101,830 [60,900, 154,000] | 152,695 [91,900, 246,000] | 0.78 [0.64, 0.88] | 1.48 [1.45, 1.67] | 65.6 [53.5, 74.7] |
| B5 | 82,481 [54,300, 126,000] | 167,604 [109,000, 232,000] | 1.13 [0.90, 1.25] | 2.75 [2.09, 2.81] | 62.7 [57.1, 69.5] |
| CGRR | 72,937 [30,400, 132,000] | 101,557 [46,500, 180,000] | 0.78 [0.64, 0.89] | 1.48 [1.45, 1.67] | 76.5 [66.1, 87.4] |
| FU | 116,613 [76,000, 166,000] | 192,859 [127,000, 294,000] | 0.78 [0.64, 0.88] | 1.48 [1.45, 1.67] | 55.6 [44.8, 63.8] |
| FC | 114,526 [71,600, 163,000] | 182,789 [119,000, 290,000] | 0.78 [0.64, 0.88] | 1.48 [1.45, 1.67] | 57.7 [46.2, 66.4] |
| A1 | 101,830 [60,900, 154,000] | 153,051 [91,900, 246,000] | 0.78 [0.64, 0.88] | 1.48 [1.45, 1.67] | 65.4 [53.5, 74.7] |
| A2a | 72,972 [30,500, 132,000] | 101,592 [46,500, 180,000] | 0.78 [0.64, 0.89] | 1.48 [1.45, 1.67] | 76.5 [66.0, 87.3] |
| A2b | 104,305 [67,300, 153,000] | 185,412 [122,000, 281,000] | 0.86 [0.71, 0.97] | 1.67 [1.45, 1.67] | 56.9 [47.1, 66.0] |
| A3 | 70,918 [29,300, 133,000] | 101,257 [45,400, 179,000] | 0.83 [0.69, 0.96] | 1.67 [1.45, 1.67] | 76.6 [66.1, 87.9] |

| Method | Missed high-risk | Relocations | Re-plans | Failed dispatches | Gini | Utilization (%) |
|---|---|---|---|---|---|---|
| B1 | 5.0 | 0 | 1 | 0 | 0.22 | 60.7 |
| B2 | 3.0 | 997 | 96 | 1 | 0.11 | 17.6 |
| B3 | 1.0 | 190 | 96 | 5 | 0.03 | 23.9 |
| B4 | 2.0 | 77 | 96 | 0 | 0.09 | 44.1 |
| B5 | 3.0 | 5,547 | 96 | 1 | 0.11 | 17.8 |
| CGRR | 1.0 | 109 | 20 | 1 | 0.06 | 35.3 |
| FU | 3.5 | 43 | 96 | 0 | 0.13 | 53.2 |
| FC | 3.0 | 63 | 96 | 0 | 0.11 | 49.5 |
| A1 | 2.0 | 76 | 21 | 0 | 0.09 | 44.3 |
| A2a | 1.0 | 109 | 96 | 1 | 0.06 | 35.4 |
| A2b | 3.0 | 0 | 1 | 24 | 0.15 | 57.1 |
| A3 | 2.0 | 75 | 21 | 19 | 0.07 | 35.3 |

## S2 Allocation results, 2019 (median [interquartile range] over 200 runs)

| Method | Worst-unit weighted unmet | Total unmet (people) | Mean response (h) | P95 response (h) | Coverage (%) |
|---|---|---|---|---|---|
| B1 | 123,695 [88,300, 166,000] | 293,336 [219,000, 387,000] | 0.94 [0.74, 1.12] | 2.08 [2.08, 2.08] | 29.5 [24.1, 35.5] |
| B2 | 122,991 [88,000, 167,000] | 301,030 [223,000, 397,000] | 1.28 [1.02, 1.46] | 3.72 [3.72, 3.72] | 27.8 [22.8, 33.7] |
| B3 | 125,563 [88,000, 170,000] | 299,584 [227,000, 396,000] | 1.14 [0.93, 1.30] | 2.08 [2.08, 2.08] | 27.6 [23.2, 33.4] |
| B4 | 125,132 [88,000, 171,000] | 299,034 [224,000, 391,000] | 2.07 [1.86, 2.25] | 12.00 [12.00, 12.00] | 28.2 [23.4, 34.0] |
| B5 | 122,991 [88,000, 167,000] | 301,110 [223,000, 397,000] | 1.28 [1.02, 1.47] | 3.72 [3.72, 3.72] | 27.8 [22.7, 33.7] |
| CGRR | 117,669 [82,600, 157,000] | 305,489 [226,000, 392,000] | 1.44 [1.16, 1.70] | 12.00 [2.08, 12.00] | 28.1 [23.0, 33.8] |
| FU | 124,541 [87,200, 169,000] | 301,700 [227,000, 396,000] | 1.23 [1.03, 1.43] | 1.73 [1.73, 2.90] | 27.4 [22.6, 32.9] |
| FC | 123,682 [86,200, 168,000] | 299,309 [223,000, 395,000] | 1.17 [1.02, 1.27] | 2.90 [2.73, 2.90] | 28.0 [23.3, 33.9] |
| A1 | 122,736 [89,000, 166,000] | 297,445 [223,000, 392,000] | 1.82 [1.64, 2.03] | 12.00 [12.00, 12.00] | 28.5 [23.7, 34.2] |
| A2a | 119,375 [84,600, 161,000] | 305,843 [226,000, 390,000] | 1.42 [1.15, 1.67] | 3.43 [2.08, 12.00] | 28.1 [23.0, 33.7] |
| A2b | 117,518 [86,400, 156,000] | 296,928 [219,000, 389,000] | 1.10 [0.90, 1.30] | 2.08 [2.08, 2.08] | 29.5 [23.3, 35.2] |
| A3 | 118,358 [86,100, 163,000] | 290,451 [215,000, 383,000] | 2.64 [1.91, 3.65] | 12.00 [12.00, 12.00] | 30.8 [24.0, 36.7] |

| Method | Missed high-risk | Relocations | Re-plans | Failed dispatches | Gini | Utilization (%) |
|---|---|---|---|---|---|---|
| B1 | 4.0 | 0 | 1 | 0 | 0.26 | 48.7 |
| B2 | 4.0 | 1,365 | 84 | 1 | 0.31 | 45.8 |
| B3 | 4.0 | 735 | 84 | 1 | 0.30 | 45.8 |
| B4 | 4.0 | 579 | 84 | 1 | 0.35 | 46.7 |
| B5 | 4.0 | 1,364 | 84 | 1 | 0.31 | 45.7 |
| CGRR | 3.0 | 391 | 69 | 0 | 0.37 | 45.7 |
| FU | 4.0 | 349 | 84 | 1 | 0.29 | 45.9 |
| FC | 3.0 | 657 | 84 | 1 | 0.29 | 46.1 |
| A1 | 4.0 | 524 | 68 | 1 | 0.34 | 47.6 |
| A2a | 4.0 | 466 | 84 | 0 | 0.36 | 45.8 |
| A2b | 3.0 | 0 | 1 | 0 | 0.46 | 46.9 |
| A3 | 3.0 | 9 | 64 | 0 | 0.40 | 49.4 |

## S3 Sensitivity to $\alpha$ and $\eta$, 2018 (CGRR, median over 200 runs)

| $\alpha$ | $\eta$ | Set coverage | Mean set size | Coverage (%) | Worst-unit unmet | Relocations | Re-plans |
|---|---|---|---|---|---|---|---|
| 0.05 | 0.00 | 0.995 | 2.73 | 84.0 | 53,618 | 181 | 96 |
| 0.05 | 0.05 | 0.995 | 2.73 | 84.0 | 53,618 | 181 | 43 |
| 0.05 | 0.10 | 0.995 | 2.73 | 84.0 | 53,618 | 181 | 41 |
| 0.05 | 0.20 | 0.995 | 2.73 | 84.0 | 53,618 | 181 | 41 |
| 0.05 | 0.30 | 0.995 | 2.73 | 84.0 | 53,618 | 181 | 41 |
| 0.10 | 0.00 | 0.990 | 2.34 | 76.5 | 72,972 | 109 | 96 |
| 0.10 | 0.05 | 0.990 | 2.34 | 76.5 | 72,937 | 109 | 33 |
| 0.10 | 0.10 | 0.990 | 2.34 | 76.5 | 72,937 | 109 | 20 |
| 0.10 | 0.20 | 0.990 | 2.34 | 76.5 | 72,937 | 109 | 14 |
| 0.10 | 0.30 | 0.990 | 2.34 | 76.5 | 72,937 | 109 | 14 |
| 0.20 | 0.00 | 0.906 | 1.80 | 59.8 | 109,592 | 59 | 96 |
| 0.20 | 0.05 | 0.906 | 1.80 | 59.9 | 109,592 | 57 | 11 |
| 0.20 | 0.10 | 0.906 | 1.80 | 59.9 | 109,592 | 55 | 8 |
| 0.20 | 0.20 | 0.906 | 1.80 | 59.9 | 109,592 | 55 | 6 |
| 0.20 | 0.30 | 0.906 | 1.80 | 59.9 | 109,592 | 55 | 6 |
| 0.30 | 0.00 | 0.779 | 1.46 | 60.1 | 109,592 | 64 | 96 |
| 0.30 | 0.05 | 0.779 | 1.46 | 60.1 | 109,592 | 63 | 27 |
| 0.30 | 0.10 | 0.779 | 1.46 | 59.4 | 109,592 | 60 | 10 |
| 0.30 | 0.20 | 0.779 | 1.46 | 59.4 | 109,592 | 60 | 7 |
| 0.30 | 0.30 | 0.779 | 1.46 | 59.4 | 109,592 | 60 | 7 |

## S3 Sensitivity to $\alpha$ and $\eta$, 2019 (CGRR, median over 200 runs)

| $\alpha$ | $\eta$ | Set coverage | Mean set size | Coverage (%) | Worst-unit unmet | Relocations | Re-plans |
|---|---|---|---|---|---|---|---|
| 0.05 | 0.00 | 0.982 | 2.91 | 30.1 | 114,358 | 262 | 84 |
| 0.05 | 0.05 | 0.982 | 2.91 | 30.2 | 114,358 | 260 | 81 |
| 0.05 | 0.10 | 0.982 | 2.91 | 30.2 | 114,448 | 242 | 66 |
| 0.05 | 0.20 | 0.982 | 2.91 | 31.1 | 114,660 | 154 | 39 |
| 0.05 | 0.30 | 0.982 | 2.91 | 31.4 | 115,351 | 128 | 34 |
| 0.10 | 0.00 | 0.975 | 2.42 | 28.1 | 119,375 | 466 | 84 |
| 0.10 | 0.05 | 0.975 | 2.42 | 28.2 | 119,305 | 464 | 82 |
| 0.10 | 0.10 | 0.975 | 2.42 | 28.1 | 117,669 | 391 | 69 |
| 0.10 | 0.20 | 0.975 | 2.42 | 28.8 | 117,304 | 277 | 55 |
| 0.10 | 0.30 | 0.975 | 2.42 | 28.2 | 119,345 | 258 | 50 |
| 0.20 | 0.00 | 0.836 | 1.79 | 26.4 | 130,172 | 381 | 84 |
| 0.20 | 0.05 | 0.836 | 1.79 | 26.4 | 130,292 | 369 | 81 |
| 0.20 | 0.10 | 0.836 | 1.79 | 26.4 | 128,932 | 301 | 60 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0.20 | 0.20 | 0.836 | 1.79 | 25.6 | 131,992 | 125 | 24 |
| 0.20 | 0.30 | 0.836 | 1.79 | 25.2 | 128,519 | 61 | 13 |
| 0.30 | 0.00 | 0.708 | 1.48 | 27.5 | 125,648 | 477 | 84 |
| 0.30 | 0.05 | 0.708 | 1.48 | 27.5 | 125,553 | 467 | 80 |
| 0.30 | 0.10 | 0.708 | 1.48 | 27.6 | 123,787 | 388 | 62 |
| 0.30 | 0.20 | 0.708 | 1.48 | 26.1 | 129,422 | 174 | 33 |
| 0.30 | 0.30 | 0.708 | 1.48 | 25.7 | 129,952 | 109 | 23 |

## S4 Coverage (%) under alternative mean need shares, 2018

| Method | 1% | 2% | 5% |
|---|---|---|---|
| B1 | 55.3 | 36.3 | 17.0 |
| B2 | 79.6 | 63.0 | 36.4 |
| B3 | 98.9 | 85.2 | 54.8 |
| B4 | 85.7 | 65.6 | 35.4 |
| B5 | 76.3 | 62.7 | 43.3 |
| CGRR | 93.9 | 76.5 | 46.0 |
| FC | 78.8 | 57.7 | 29.7 |
| FU | 75.6 | 55.6 | 27.8 |

## S4 Coverage (%) under alternative mean need shares, 2019

| Method | 1% | 2% | 5% |
|---|---|---|---|
| B1 | 47.7 | 29.5 | 13.1 |
| B2 | 45.2 | 27.8 | 12.3 |
| B3 | 43.7 | 27.6 | 12.6 |
| B4 | 43.6 | 28.2 | 12.9 |
| B5 | 45.2 | 27.8 | 12.3 |
| CGRR | 44.9 | 28.1 | 12.4 |
| FC | 44.5 | 28.0 | 12.6 |
| FU | 43.5 | 27.4 | 12.4 |

## S5 Run log


```
2026-11-06 16:06:06 START smoke_test=False python=3.13.18 numpy=2.4.6 pandas=2.2.3
2026-11-06 16:06:06 CONFIG {"smoke_test": false, "events": ["2018", "2019"], "data_dir": "/content/drive/MyDrive/Paper4_Kerala", "results_dir":
"/content/drive/MyDrive/Paper4_Kerala/results", "model": {"alpha": 0.1, "eta": 0.1, "beta": 0.5, "tau_max": 12.0, "lam": 0.01, "mu": 0.05, "rho": 0.5, "fixed_width": 0.3,
"b4_gamma_frac": 0.5, "need_beta": [2.0, 98.0], "resources": {"team": {"kappa": 150, "hmax": 0.3, "mode": "road"}, "boat": {"kappa": 60, "hmax": 99.0, "mode": "boat"}, "medical":
{"kappa": 80, "hmax": 0.3, "mode": "road"}}}, "experiment": {"n_runs": 200, "n_scen_b3": 30, "b5_search": 60, "seed": 42, "sensitivity_alpha": [0.05, 0.1, 0.2, 0.3],
"sensitivity_eta": [0.0, 0.05, 0.1, 0.2, 0.3], "demand_means": [0.01, 0.02, 0.05]}}
2026-11-06 16:06:06 [2018] temperature T=1.425  qhat=0.875  delta=[7.00e-06 7.60e-05 2.63e-04 4.61e-04]  Gamma=6
```

```
2026-11-06 16:06:06 [2018] test ECE raw=0.107 -> scaled=0.150; Brier 0.470 -> 0.486
2026-11-06 16:06:10   B1    replans=  1  coverage=36.3%
2026-11-06 16:06:11   B2    replans= 96  coverage=63.0%
2026-11-06 16:06:37   B3    replans= 96  coverage=85.2%
2026-11-06 16:06:45   B4    replans= 96  coverage=65.6%
2026-11-06 16:06:45   B5    replans= 96  coverage=62.7%
2026-11-06 16:06:47   CGRR  replans= 20  coverage=76.5%
2026-11-06 16:06:50   A1    replans= 21  coverage=65.4%
2026-11-06 16:06:55   A2a   replans= 96  coverage=76.5%
2026-11-06 16:06:55   A2b   replans=  1  coverage=56.9%
2026-11-06 16:06:59   A3    replans= 21  coverage=76.6%
2026-11-06 16:07:05   FU    replans= 96  coverage=55.6%
2026-11-06 16:07:09   FC    replans= 96  coverage=57.7%
2026-11-06 16:43:14   S1: best baseline B3
2026-11-06 16:43:14   S2: best baseline B3
2026-11-06 16:43:14   S3: best baseline B3
2026-11-06 16:43:14   S4: best baseline B3
2026-11-06 16:43:37   alpha=0.05 done
2026-11-06 16:43:49   alpha=0.1 done
2026-11-06 16:43:58   alpha=0.2 done
2026-11-06 16:44:09   alpha=0.3 done
2026-11-06 16:44:10 [2018] done
2026-11-06 16:44:10 [2019] temperature T=1.425  qhat=0.875  delta=[7.00e-06 7.60e-05 2.63e-04 4.61e-04]  Gamma=6
2026-11-06 16:44:10 [2019] test ECE raw=0.169 -> scaled=0.151; Brier 0.560 -> 0.556
2026-11-06 16:44:12   B1    replans=  1  coverage=29.5%
2026-11-06 16:44:12   B2    replans= 84  coverage=27.8%
2026-11-06 16:53:40   B3    replans= 84  coverage=27.6%
2026-11-06 17:19:47   B4    replans= 84  coverage=28.2%
2026-11-06 17:19:47   B5    replans= 84  coverage=27.8%
2026-11-06 17:21:23   CGRR  replans= 69  coverage=28.1%
2026-11-06 17:44:01   A1    replans= 68  coverage=28.5%
2026-11-06 17:45:43   A2a   replans= 84  coverage=28.1%
2026-11-06 17:45:44   A2b   replans=  1  coverage=29.5%
2026-11-06 17:45:48   A3    replans= 64  coverage=30.8%
2026-11-06 17:46:05   FU    replans= 84  coverage=27.4%
2026-11-06 17:48:44   FC    replans= 84  coverage=28.0%
2026-11-06 18:05:02   S1: best baseline B1
2026-11-06 18:05:02   S2: best baseline B1
2026-11-06 18:05:02   S3: best baseline B1
2026-11-06 18:05:02   S4: best baseline B1
2026-11-06 18:06:22   alpha=0.05 done
2026-11-06 18:13:20   alpha=0.1 done
2026-11-06 18:19:52   alpha=0.2 done
2026-11-06 18:25:06   alpha=0.3 done
2026-11-06 18:25:07 [2019] done
2026-11-06 18:25:07 END
INPUT_SHA256_16 {"2018": {"forecasts": "1dec4a31025bb1cb", "stocks": "610e0187999926bd", "route_times": "3c0c78b75c9cb3c4", "exposure": "1141856b81f400e1", "tau0":
"2be6eec7bf6fe711", "calibration": "49a98d2fdc626456"}, "2019": {"forecasts": "bd4d8ca4f91c3092", "stocks": "e8d562e6fa3e1c44", "route_times": "f3aff08bfb6d8387", "exposure":
"1141856b81f400e1", "tau0": "2be6eec7bf6fe711", "calibration": "49a98d2fdc626456"}}
```